

Rate-Aware Decentralized Detection with Many Sensors

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Abstract—[Abstract placeholder.]

Index Terms—decentralized detection, sensor networks, error exponent, Chernoff information, communication rate

I. Introduction

[Introduction placeholder.]

II. Problem Formulation

Consider binary hypotheses H_1 and H_2 , and let $H^* \in \{1, 2\}$ index the true one, with priors $P(H^* = 1) = p$ and $P(H^* = 2) = 1 - p$. Sensor $i \in \{1, \dots, N\}$ observes $y^i \in \mathcal{Y}$ and applies an encoder $\gamma^i : \mathcal{Y} \rightarrow \mathcal{U}$. Its message $u^i = \gamma^i(y^i)$ is available to a fusion center (FC), which uses $\gamma^0 : \mathcal{U}^N \rightarrow \{1, 2\}$ to produce the final decision $u^0 = \gamma^0(u^{1:N})$.

Under uniform error cost the Bayes risk is the probability of error,

$$J^N(\gamma^{0:N}) \triangleq \mathbb{P}^{\gamma^{0:N}}(u^0 \neq H^*) = P_e \quad (1)$$

We use the positive normalized error exponent

$$J_{EE}^N(\gamma^{0:N}) \triangleq -\frac{1}{N} \log J^N(\gamma^{0:N}), \quad (2)$$

which is maximized when the error probability decays rapidly.

A. Assumptions

The analysis uses the following assumptions.

- 1) The observations $y^1, \dots, y^N$ are independent and identically distributed conditioned on the true hypothesis.
- 2) Under both hypotheses, the observation laws have densities $f_{y|H}(y|H_j)$ with respect to a common reference measure.
- 3) All sensors have the same finite message alphabet $\mathcal{U}$.
- 4) The likelihood ratio

$$L(y) := \frac{f_{y|H}(y|H_2)}{f_{y|H}(y|H_1)}$$

is regular enough that threshold encoders are well defined; in particular, its derivative is continuous and nonsingular almost surely on the observation space.

- 5) The FC knows the sensor encoders, including any realized common randomization, as well as the received messages.

Under these conditions, the likelihood ratio is a sufficient statistic at each sensor. The results of [1] guarantee

an optimal finite-population solution with threshold encoders and a maximum a posteriori (MAP) decoder. They also show that identical independent sensor policies are optimal in the infinite-population problem and asymptotically optimal for large finite populations.

B. MAP fusion rule

For fixed deterministic encoders, define

$$\Delta_N \triangleq \frac{1}{N} \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \quad (3)$$

The MAP rule is

$$\gamma^{0*}(u^{1:N}) = \begin{cases} H_1 & \text{if } \Delta_N \geq t \\ H_2 & \text{if } \Delta_N < t \end{cases} \quad t = \frac{1}{N} \log \frac{1-p}{p} \quad (4)$$

III. Error-Exponent Characterization

For a sensor policy γ^i , define the policy-dependent Chernoff coefficient

$$M(\alpha, \gamma^i) \triangleq \sum_{u \in \mathcal{U}} P^{\gamma^i}(u|H_2)^{1-\alpha} P^{\gamma^i}(u|H_1)^\alpha, \quad 0 \leq \alpha \leq 1. \quad (5)$$

Applying Chernoff bounds to the two MAP error events gives

$$P_e \leq 2p^\alpha(1-p)^{1-\alpha} \prod_{i=1}^N M(\alpha, \gamma^i) \quad (6)$$

When every sensor uses the same policy γ , taking $-\frac{1}{N} \log(\cdot)$ of (6) and letting $N \rightarrow \infty$ leaves the Chernoff information between the two induced message distributions,

$$\liminf_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq -\min_{\alpha \in [0,1]} \log M(\alpha, \gamma) \quad (7)$$

The structural result in [1] therefore reduces the large-network design to selecting one encoder that maximizes this exponent, subject here to a communication-rate constraint.

IV. Rate-Aware R -Bit Threshold Policies

Assume

$$y^i|H_1 \sim \mathcal{N}(0, \sigma^2), \quad y^i|H_2 \sim \mathcal{N}(\mu, \sigma^2), \quad (8)$$

with $\mu > 0$. Then

$$\log L(y) = \frac{\mu y}{\sigma^2} - \frac{\mu^2}{2\sigma^2} \quad (9)$$

We generalize the always-transmit policy and the confidence-based silent policy into a single family of threshold policies, indexed by the number of bits R , a bin width $\Delta > 0$, and a silence half-width $\delta \geq 0$.

A. The R -bit ladder

Let $L \triangleq 2^R$ and $\mathcal{U} = \{u^1, \dots, u^L, u^S\}$, where u^S denotes silence. For $\delta \geq 0$, define the thresholds

$$t_j(\delta) \triangleq \begin{cases} \frac{\mu}{2} - \delta - \left(\frac{L}{2} - j\right)\Delta, & j = 1, \dots, \frac{L}{2} \\ \frac{\mu}{2} + \delta + \left(j - \frac{L}{2} - 1\right)\Delta, & j = \frac{L}{2} + 1, \dots, L \end{cases} \quad (10)$$

with $t_0 \triangleq -\infty$ and $t_{L+1} \triangleq +\infty$, so the thresholds cut the line into $L + 1$ intervals (t_{j-1}, t_j) , $j = 1, \dots, L + 1$. The sensor sends u^j from the j -th interval for $j \leq L/2$, stays silent on the central one, $j = L/2 + 1$, and sends u^{j-1} for $j \geq L/2 + 2$.

At $\delta = 0$ the two central thresholds coincide at $\mu/2$: the silence interval degenerates to a point, u^S is never sent, and what remains is an L -region always-transmit policy. We write γ^A for the policy at $\delta = 0$, and γ^S for the policy at $\delta > 0$.

B. Symbol probabilities

Standardizing by σ , write

$$a_j \triangleq \frac{t_j(\delta)}{\sigma}, \quad s \triangleq \frac{\mu}{\sigma},$$

with $a_0 \triangleq -\infty$ and $a_{L+1} \triangleq +\infty$. Since $\log L(y)$ in (9) is increasing in y , every decision region is an interval and its probability is a difference of Q -functions,

$$\begin{aligned} P(t_{j-1} < y < t_j \mid H_1) &= Q(a_{j-1}) - Q(a_j), \\ P(t_{j-1} < y < t_j \mid H_2) &= Q(a_{j-1} - s) - Q(a_j - s) \end{aligned}$$

Every quantity below is written in terms of a_j and s , and applies to γ^A and γ^S alike.

C. Symmetry and the closed-form bound

The thresholds (10) mirror about $\mu/2$, that is $t_{L+1-j}(\delta) = \mu - t_j(\delta)$, and the two densities mirror with them, $f_{y|H}(y|H_1) = f_{y|H}(\mu - y|H_2)$. The message probabilities are therefore exchanged by the mirror, $P(u^m|H_1) = P(u^{L+1-m}|H_2)$, and the silence probability is hypothesis-independent,

$$\begin{aligned} P(u^S|H_1) &= P(u^S|H_2) \triangleq P(u^S) \\ &= Q\left(\frac{s}{2} - \frac{\delta}{\sigma}\right) - Q\left(\frac{s}{2} + \frac{\delta}{\sigma}\right) \end{aligned} \quad (11)$$

Generalizing (5) to the $L + 1$ symbols, the silence term collapses for every α by (11):

$$M(\alpha; R, \Delta, \delta) \triangleq P(u^S) + \sum_{m=1}^L P(u^m|H_2)^{1-\alpha} P(u^m|H_1)^\alpha$$

The mirror relation gives $M(1 - \alpha; \cdot) = M(\alpha; \cdot)$, and each summand is convex in α , so M is convex on $[0, 1]$ and

symmetric about $\alpha = 1/2$: its minimum sits there. With (7), this gives the bound in closed form,

$$\begin{aligned} C(R, \Delta, \delta) &\triangleq -\log M(0.5; R, \Delta, \delta) \\ &= -\log \left[P(u^S) + \sum_{m=1}^L \sqrt{P(u^m|H_1)P(u^m|H_2)} \right] \end{aligned} \quad (12)$$

$$\liminf_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq C(R, \Delta, \delta) \quad (13)$$

At $\delta = 0$ we have $P(u^S) = 0$ and (12) reduces to the L -region always-transmit bound $C_A \triangleq C(R, \Delta, 0)$; for $\delta > 0$ we write $C_S \triangleq C(R, \Delta, \delta)$.

D. The unquantized ceiling

Before evaluating the bound it is worth fixing what it is bounded by. Suppose the sensor forwards y^i itself, at infinite precision. Under the Gaussian model (8) its Chernoff coefficient is $\exp(-\alpha(1-\alpha)s^2/2)$, again symmetric about $\alpha = 1/2$ and convex, so its minimum sits there as in (12), giving

$$\begin{aligned} C_\infty &\triangleq -\min_{\alpha \in [0,1]} \log \int f_{y|H}^{1-\alpha}(y|H_2) f_{y|H}^\alpha(y|H_1) dy \\ &= \frac{\mu^2}{8\sigma^2} = \frac{s^2}{8} \end{aligned} \quad (14)$$

Since $H^* \rightarrow y^i \rightarrow u^i$ is Markov for any policy, the data processing inequality gives $I(H^*; u^i) \leq I(H^*; y^i)$: quantization can discard evidence about the hypothesis, never create it. Hence

$$C(R, \Delta, \delta) \leq C_\infty \quad \text{for every } R, \Delta, \delta, \quad (15)$$

and C_∞ is the natural yardstick against which every finite-rate policy is measured. Note that C_∞ depends on the model only through $s = \mu/\sigma$: the whole problem is how many standard deviations apart the two hypotheses sit.

E. Transmitted rate and the design problem

The parameter R is the rate γ^A pays, but not the rate γ^S pays: a silent sensor sends nothing. Using (11), the expected transmitted rate is

$$\bar{R} \triangleq R(1 - P(u^S)), \quad (16)$$

so $\bar{R} = R$ for γ^A and $\bar{R} < R$ for γ^S . Comparisons made at equal R therefore charge γ^S for bandwidth it never uses.

At a given rate the design problem is to place the ladder,

$$(\Delta^*, \delta^*) = \arg \max_{\Delta > 0, \delta \geq 0} C(R, \Delta, \delta), \quad (17)$$

with γ^A obtained by pinning $\delta = 0$ and optimizing over Δ alone. For $R > 1$ this admits no closed form, and we solve it numerically.

TABLE I

Optimal ladder parameters of γ^S against R , in units of σ , at SNR = 0 dB. Obtained by maximizing $C(R, \Delta, \delta)$ directly, so no simulation is involved and the range extends past the swept $R \leq 5$. At $R = 1$ the single pair of thresholds is $\mu/2 \mp \delta$ and Δ is inert.

R	2	3	4	5	6	7
Δ^*	0.867	0.529	0.310	0.176	0.099	0.054
δ^*	0.384	0.224	0.127	0.070	0.038	0.021
δ^*/Δ^*	0.443	0.424	0.409	0.396	0.387	0.379
$\Delta^*(R)/\Delta^*(R-1)$	—	0.610	0.585	0.570	0.559	0.551

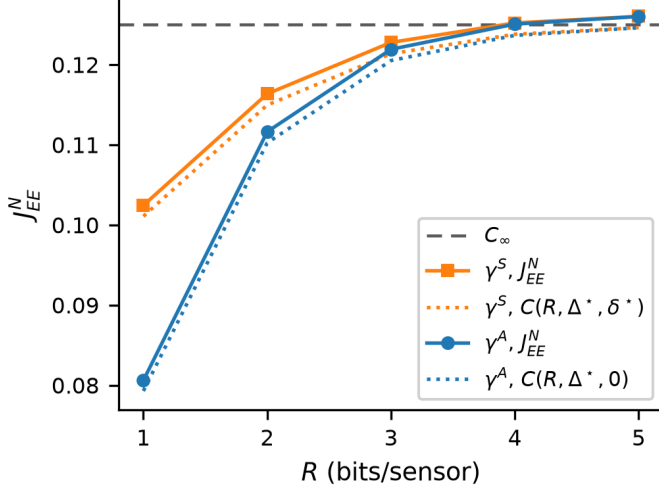


Fig. 1. Normalized error exponent J_{EE}^N against R at the optimal (Δ^*, δ^*) , for $N = 3000$ and SNR = 0 dB, with the closed-form bounds of (12) (dotted) and the unquantized ceiling $C_\infty = 0.125$ of (14) (dashed).

V. Numerical Results

We evaluate the ladder for $R = 1, \dots, 5$ at $N = 3000$ sensors and $p = \frac{1}{2}$, at three operating points SNR $\in \{-5, 0, +5\}$ dB. At each point (Δ^*, δ^*) solves (17). The fusion center computes J^N exactly, from which J_{EE}^N follows by (2).

Figure 1 shows the 0 dB panel, and Figures 2 and 3 the other two operating points; one shape repeats at all three. J_{EE}^N rises monotonically in R and approaches the unquantized ceiling C_∞ of (14). It sits slightly above its own bound throughout – by 0.0012 to 0.0016 across the three panels at $N = 3000$, essentially flat in R – since C bounds the $N \rightarrow \infty$ limit while the finite- N exponent still carries an $O(1/N)$ prefactor. The same offset carries it a little past C_∞ itself at high R , most visibly at -5 dB where the ceiling is smallest: (15) constrains the limit, not the finite- N exponent.

Saturation is fast: a single bit already recovers 63% of C_∞ under γ^A and 81% under γ^S , and by $R = 3$ both policies sit within a few percent of the ceiling. The entire design question therefore lives at $R \leq 3$; past that the extra bits buy nothing.

The silent policy dominates the always-transmit one at every R , as $C_S > C_A$ predicts, but the whole of that

advantage is concentrated at low rate. It is largest at $R = 1$ and by $R = 4$ the two curves are indistinguishable. The optimized geometry explains why. Table I lists (Δ^*, δ^*) against R : the cell width Δ^* decays geometrically, settling toward a factor of about 0.55 per added bit, while the ratio δ^*/Δ^* stays close to 0.4 throughout. The silence cell holds an essentially fixed fraction of a cell, so it shrinks with the cell; by (11) the silence probability vanishes with it, and γ^S degenerates continuously into γ^A . The two effects visible at the knee of Figure 1 – silence ceasing to help, and quantization ceasing to cost – are the same phenomenon, $\Delta^* \rightarrow 0$, seen from two sides.

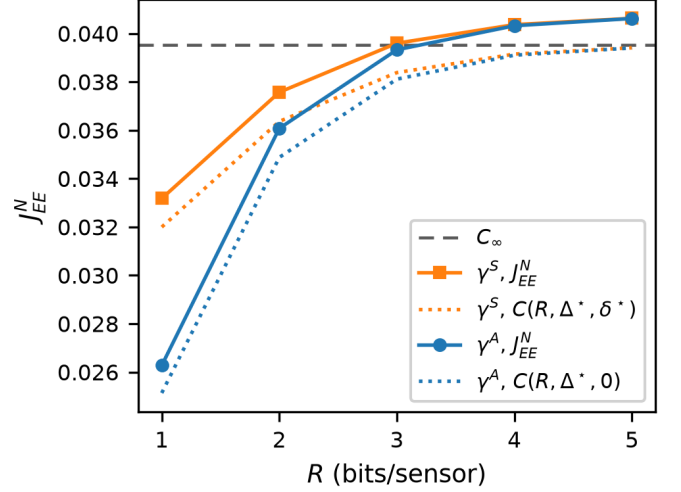


Fig. 2. As Figure 1, at SNR = -5 dB, where $C_\infty \approx 0.0395$.

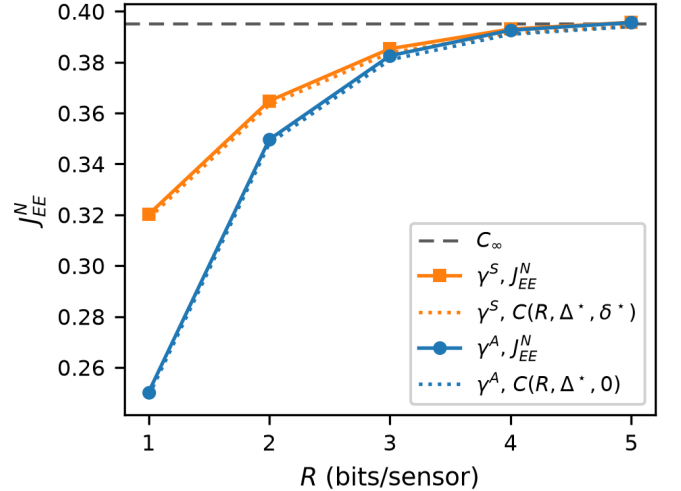


Fig. 3. As Figure 1, at SNR = $+5$ dB, where $C_\infty \approx 0.3953$.

Across the three operating points only the vertical scale changes. The ceiling moves from $C_\infty \approx 0.0395$ at -5 dB to 0.125 at 0 dB to 0.3953 at $+5$ dB, an order of magnitude, while the shape of both curves is preserved: normalized by C_∞ the bound curves agree to within 0.7 percentage

points, and (Δ^*, δ^*) in units of σ barely move. This is (14) showing through – the model enters only through s , which C_∞ absorbs. SNR sets the achievable exponent; R sets what fraction of it is achieved.

Both policies are charged the nominal R , which by (16) overstates what γ^S sends: at $R = 1$ and 0 dB it transmits $\bar{R} \approx 0.59$ bits per sensor, below the one bit γ^A must spend.

VI. Conclusion

We combined the optimal-policy results for large sensor populations with a rate-aware encoder, generalizing the one-bit always-transmit and silent policies into a single R -bit threshold family whose error exponent is available in closed form. Measured against the unquantized ceiling $C_\infty = s^2/8$, the family saturates quickly: one bit per sensor already buys 63% of C_∞ without silence and 81% with it, three bits come within a few percent, and beyond that the extra rate is wasted. Selective silence is therefore worth the most at exactly the rate where rate is scarcest – at $R = 1$ it raises the exponent while transmitting $\bar{R} \approx 0.59$ bits per sensor, where the always-transmit policy must spend a full bit – and its advantage vanishes as the optimal cell width shrinks and the silence cell closes with it.

References

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